

4-2 Assignment: Hash Tables

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The code aims to implement a hash table using a singly linked list to manage bids loaded from a CSV file. Its primary purpose is to provide functionalities like inserting bids, displaying all bids, finding bids by ID, removing bids, and exiting the program. Overall, developing this code was a valuable learning experience that enhanced my understanding of data structures and algorithms, particularly in the context of implementing a hash table with linked lists. It provided insights into effective problem-solving strategies and reinforced the importance of careful planning and attention to detail in software development.

Pseudocode:

Start program

Create a structure to hold bid information (e.g., Bid)

Create a structure for linked list nodes (e.g., Node)

Create a class for the hash table (e.g., HashTable)

Initialize hash table size and linked list array

Implement destructor to free memory allocated for linked lists

Implement hash function to calculate key from bid ID

Implement method to insert a bid into the hash table

Calculate key for the bid

Check if a linked list exists at the calculated key index

If not, create a new linked list and add the bid to it

If yes, add the bid to the end of the existing linked list

Implement method to print all bids in the hash table

Iterate through each linked list in the array

Print all bids in each linked list

Implement method to remove a bid from the hash table

Calculate key for the bid to be removed

Search for the bid in the linked list at the calculated key index

If found, remove the bid from the linked list

Implement method to search for a bid in the hash table

Calculate key for the bid to be searched

Search for the bid in the linked list at the calculated key index

If found, return the bid

End program